

Spectrum-Conditioned DE and Kolmogorov-Arnold Quantum Residual Learning for Cross-Subject EEG Emotion Recognition

Abstract—Cross-subject electroencephalography (EEG) emotion recognition requires representations that retain affective information despite variation between individuals. We propose QuanKAN, a hybrid quantum-classical framework that coordinates input conditioning, subject-generalizable representation learning, and selective quantum residual refinement. A lightweight encoder extracts short-time spectral context from differential-entropy features smoothed by a linear dynamical system (DE-LDS) and uses it to condition electrode- and band-specific activity before graph-based spatial encoding and temporal aggregation. Subject-adversarial, supervised contrastive, and prototype objectives regularize the shared representation. A Kolmogorov-Arnold network (KAN) interface maps a detached embedding through a four-qubit, four-layer parameterized quantum circuit, whose features enter the classical pathway through a predictive-entropy gate and a bounded fusion coefficient. Under strict leave-one-subject-out evaluation, QuanKAN achieves accuracy/F1 scores of $87.41 \pm 6.05\%/87.12 \pm 6.64\%$, $78.43 \pm 4.48\%/78.12 \pm 4.67\%$, $86.81 \pm 3.72\%/86.60 \pm 3.95\%$, and $73.81 \pm 4.58\%/73.16 \pm 4.91\%$ on SEED, SEED IV, SEED V, and SEED VII, respectively. The model contains 589,366–691,652 parameters across these datasets. Ablations favor the complete architecture over its component-removal and alternative-head variants, including a capacity-matched classical residual. These results support selective quantum residual learning within a compact EEG representation model; the circuit is evaluated by state-vector simulation.

Index Terms—EEG emotion recognition, cross-subject learning, differential entropy, graph neural networks, Kolmogorov-Arnold networks, quantum machine learning, hybrid quantum-classical learning.

I. INTRODUCTION

Electroencephalography (EEG) provides a non-invasive measure of neural activity with high temporal resolution, making it useful for affective brain-computer interfaces. Emotion-related EEG patterns nevertheless vary across individuals because of differences in neural baselines, temporal dynamics, and the spatial distribution of band-specific activity [1]–[3]. A model that performs well on samples from familiar participants may therefore generalize poorly to an unseen participant. Differential-entropy (DE) features offer compact, band-specific representations, while channel- and connectivity-aware processing helps capture their spatial organization [4]–[6]. The central challenge is to preserve affective structure while limiting reliance on subject-specific patterns.

Graph-based encoders and temporal aggregation address the spatial and temporal organization of EEG. Domain adaptation, representation alignment, and source-aware learning further

address variation between participants [7]–[10]. These directions motivate a coordinated design: input refinement should support the downstream encoder, and regularization should organize its representation across source subjects. In particular, the temporal variation of precomputed DE features provides a contextual signal that can guide electrode- and band-specific processing before spatial aggregation.

Kolmogorov-Arnold networks (KANs) parameterize non-linear mappings through learnable univariate functions and have been investigated in graph learning, temporal modeling, scientific machine learning, and hardware implementations [11]–[14]. Parameterized quantum circuits (PQCs) provide another trainable feature transformation, whose behavior depends on data encoding, circuit structure, optimization, and measured observables [15]–[17]. Incorporating these mappings into an EEG model raises a concrete design question: how should a compact quantum branch interact with a classical representation that already supports emotion classification?

QuanKAN addresses this question through selective residual refinement. Short-time spectral context from DE features smoothed by a linear dynamical system (DE-LDS) first conditions the input. A graph-aware spatio-temporal encoder then learns a shared embedding under subject-adversarial, supervised contrastive, and prototype objectives. The quantum branch transforms a gradient-detached copy of this embedding through KAN-based angle encoding, data re-uploading, and measurement readout. Its feature residual is scaled by detached classical predictive entropy and a bounded, learned coefficient before hybrid classification. This design assigns the quantum pathway a specific role within the representation learner, which can be examined through component ablations and classical replacement controls.

The main contributions are as follows:

- **Spectrum-conditioned DE-LDS refinement** that applies bounded electrode-band modulation, a context shift, and temporal-graph residual corrections before encoding.
- **Graph-aware spatio-temporal encoding** that combines graph attention, learned spatial tokens, residual temporal convolution, and attention pooling.
- **Multi-level cross-subject regularization** that coordinates subject-adversarial learning, supervised contrastive learning, and subject-emotion prototype alignment.
- **A KAN-based classical-quantum interface** that parameterizes classification, circuit-angle encoding, and measurement readout with adaptive spline mappings.

- **Uncertainty-gated quantum residual learning** that selectively injects data-reuploading features using detached encoding, predictive-entropy gating, and a bounded fusion coefficient.

The following sections position the framework relative to existing methods, formulate its architecture and objectives, describe strict leave-one-subject-out evaluation, and analyze predictive performance, component contributions, and computational cost.

II. RELATED WORK

A. Spectral and Spatio-Temporal EEG Representation Learning

Recent EEG emotion recognition models combine spectral representations with graph-based spatial processing. Self-supervised graph attention, convolutional and adaptive spatio-temporal graph networks, and learned channel-importance mechanisms model relationships between electrodes [18]–[20]. Graph-based feature purification and region-dependent temporal modeling further address heterogeneity in EEG representations [21], [22]. Connectivity-based convolution, convolutional-transformer fusion, and channel-wise processing provide related architectural context from EEG decoding tasks beyond emotion recognition [23]–[25]. Within emotion recognition, temporal-spectral-spatial attention and sub-band embedding offer additional ways to combine feature domains [26], [27].

QuanKAN builds on this spatial and temporal perspective by conditioning the input before graph encoding. Its short-time spectrum is computed from electrode-averaged DE-LDS trajectories. The resulting context describes variation in the precomputed feature streams; it is distinct from the spectrum of the raw EEG waveform. Bounded gains and an additive shift connect this context directly to the electrode-band representation processed by the encoder.

B. Cross-Subject Learning and Representation Alignment

Cross-subject EEG methods investigate contrastive learning, identity-invariant representations, and attention-based refinement [28]–[30]. Semi-supervised alignment, prompt-guided generalization, source-reliability assessment, and adversarial transfer provide further strategies for handling participant variation [31]–[34]. Information-bottleneck models, class-wise alignment, and graph-matching domain adaptation likewise organize representations across domains [35]–[37].

These methods must be interpreted in relation to target-data access. Adaptation settings that use target-subject observations during training differ from strict evaluation on an entirely held-out participant. QuanKAN uses source-subject labels to coordinate three objectives: subject-adversarial learning opposes subject prediction, supervised contrastive learning aligns samples with the same emotion label, and prototype alignment relates subject-emotion centroids to class-level centroids. Their roles are complementary by construction, while their empirical contributions are assessed through held-out-subject ablations.

C. KAN Mappings and Hybrid Quantum-Classical Learning

KANs construct nonlinear mappings from learnable univariate functions. QuanKAN uses a KAN-linear operator with a SiLU base term and cubic B-spline expansion; separate instances implement classical classification, quantum-angle encoding, measurement readout, and hybrid classification. This explicit parameterization connects the classical embedding to the circuit and its observable features.

Hybrid quantum-classical models combine classical feature processing with PQC and classical readout. Repeated encoding and observable selection determine how the circuit transforms its inputs. QuanKAN uses a four-qubit data-reuploading circuit to generate a feature residual from a detached embedding. A separate auxiliary head predicts the signed discrepancy between the emotion label and classical class probabilities. Detached predictive entropy controls residual injection, but is not treated as a calibrated error probability. The resulting methodological focus is the coordination of spectral conditioning, cross-subject regularization, and selective residual fusion. Evaluation uses state-vector simulation and does not establish a quantum computational advantage.

III. METHODOLOGY

A. Architecture Overview

Let a batch of precomputed DE-LDS features be represented as

$$\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F}, \quad (1)$$

Here B is the batch size, T is the number of temporal windows, $N = 62$ is the number of electrodes, and $F = 5$ is the number of frequency bands. The indices i, t, n, f identify the sample, window, electrode, and band, respectively. We write $X_{i,t,n,f}$ for a scalar entry and $\mathbf{X}_{i,t} = \mathbf{X}_{i,t,:,:} \in \mathbb{R}^{N \times F}$ for a sample-window slice, and use the same convention for subsequent feature tensors. The emotion label is $y_i \in \{1, \dots, C\}$, with $C \geq 2$, and the subject label is $s_i \in \{1, \dots, N_s\}$, where N_s is the number of training subjects. Subject labels are used only during training. Fig. 1 shows the six stages of QuanKAN. Precomputed DE-LDS features (1) pass through the Feature Enhancer (2), which preserves the input dimensions, and the Spatio-Temporal Encoder (3). Layer normalization (LN), a fully connected projection, and the Gaussian error linear unit (GELU) produce the shared 192-dimensional embedding (4). This embedding feeds the Hybrid Classifier (5) and the prototype, contrastive, and subject-adversarial heads (A–C). Within the Hybrid Classifier, the classical prediction determines the entropy gate, while a detached copy of the embedding enters the quantum branch. The gated quantum feature residual is added to the classical feature before the final emotion prediction (6). The Frontend Head (D) supervises the spectral context, and the Quantum Head (E) provides auxiliary residual prediction. Raw EEG acquisition and modality preparation precede the model, whose input consists of precomputed DE-LDS features.

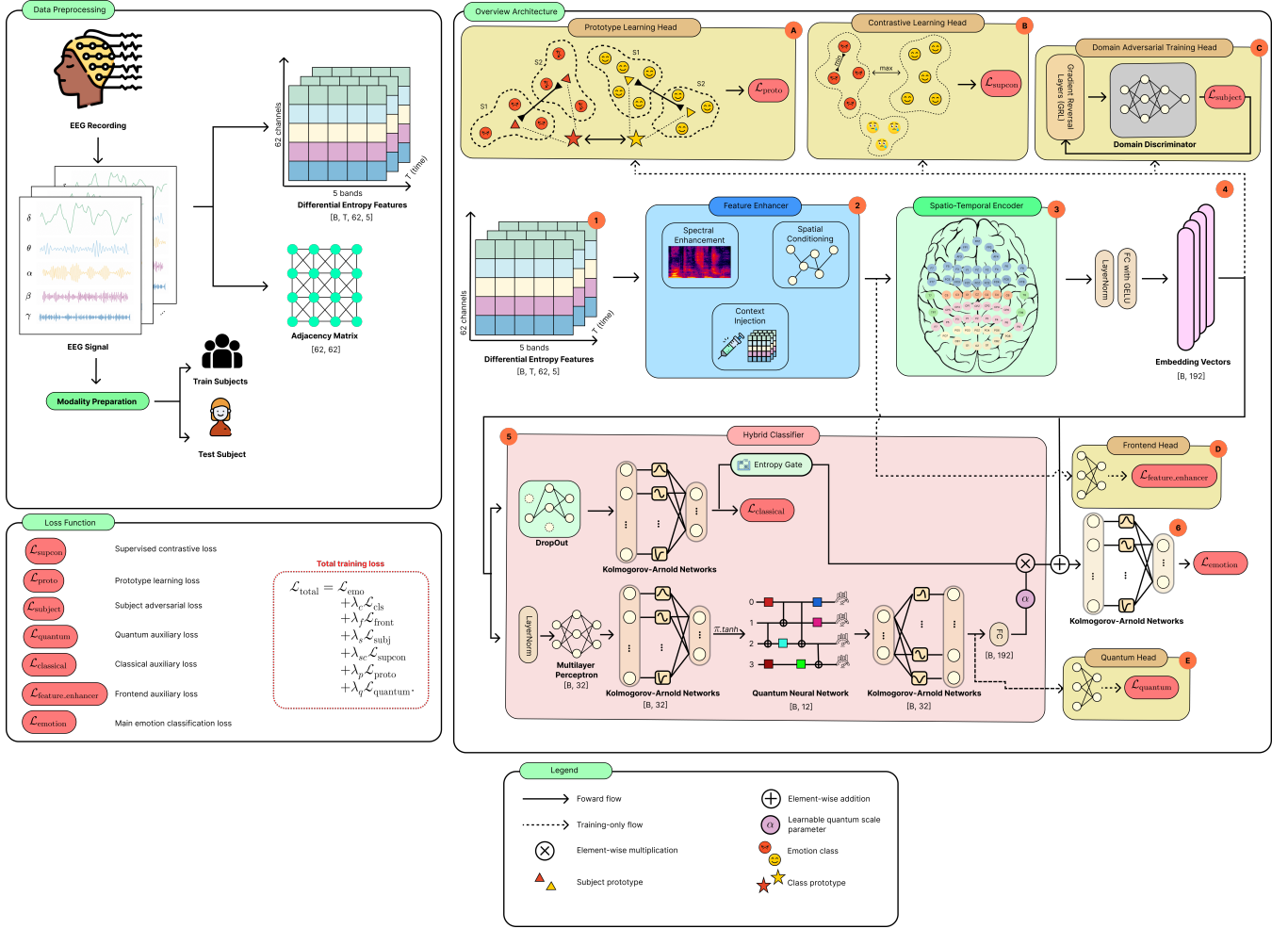


Fig. 1. Overview of QuanKAN. The main pathway (1–6) processes precomputed DE-LDS features through spectral conditioning, spatio-temporal encoding, and hybrid classification. Heads A–C regularize the shared embedding through prototype, contrastive, and subject-adversarial objectives. The classical prediction supplies an entropy gate that scales the quantum feature residual together with a bounded fusion coefficient. Head D supervises the 64-dimensional spectral context, and Head E predicts a classwise residual from the 32-dimensional quantum representation.

B. Feature Enhancer: Spectrum-Conditioned DE Refinement

The Feature Enhancer in Fig. 2 first extracts spectral context from the DE-LDS streams and uses it to modulate the input. Temporal filtering and a fixed-graph operator then produce residual corrections from the conditioned features. These corrections are combined with the conditioned input, and a validity mask is applied to the resulting sequence.

1) *Spectral enhancement: validity masking and context encoding*: All-zero temporal windows are treated as padding. The indicator $\mathbb{I}$ equals one when its argument holds and zero otherwise. For sample i and window t , the binary validity mask $M_{i,t} \in \{0, 1\}$ and stabilized valid-window count d_i are defined as

$$M_{i,t} = \mathbb{I} \left(\sum_{n=1}^N \sum_{f=1}^F |X_{i,t,n,f}| > 0 \right), \quad (2)$$

$$d_i = \max \left(1, \sum_{t=1}^T M_{i,t} \right).$$

The electrode-averaged DE stream for each frequency band is

$$U_{i,f,t} = \frac{1}{N} \sum_{n=1}^N X_{i,t,n,f}. \quad (3)$$

Each band stream $U_{i,f,:}$ is standardized using masked temporal means and population variances, with d_i as the denominator and $\epsilon_x = 10^{-4}$ for numerical stabilization. Invalid positions are set to zero, giving $\hat{U}_{i,f,t}$. The normalized multiband sequence is $\hat{\mathbf{U}}_i = [\hat{U}_{i,f,t}]_{f,t} \in \mathbb{R}^{F \times T}$. These operations act on precomputed DE-LDS features. A sequence containing only padded windows yields a zero normalized stream.

2) *Short-time spectrum and context encoder*: A length-three moving average smooths $\hat{\mathbf{U}}$ before a centered short-time Fourier transform (STFT) with zero padding, a 16-sample periodic Hann window, and a hop size of 4. The log-power representation is then standardized:

$$\tilde{\mathbf{S}}_i = \text{Std}_{\nu,\tau} \left[\log \left(1 + \left| \text{STFT}(\text{MA}_3(\hat{\mathbf{U}}_i)) \right|^2 \right) \right], \quad (4)$$

Here MA_3 is the length-three moving average, and STFT uses the configuration specified above. The indices ν and τ denote frequency bins and frames. The operator $\text{Std}_{\nu,\tau}$ standardizes each sample-band plane over these dimensions using ϵ_x . The STFT bins describe temporal variation in the DE streams, while the five original EEG bands remain separate channels. A lightweight MobileNetV3-based encoder processes this representation. Its stem maps 5 to 16 channels using a 3×3 convolution. The inverted residual blocks use $(c_{\text{in}}, c_{\text{exp}}, c_{\text{out}}, s)$ settings of $(16, 48, 24, 1)$, $(24, 72, 32, 2)$, $(32, 96, 48, 1)$, $(48, 144, 64, 2)$, and $(64, 192, 64, 1)$. Batch normalization and hard-swish follow pointwise expansion and depthwise 3×3 convolution. Squeeze-and-excitation recalibrates the channels before pointwise projection, and residual connections are used where the stride and channel dimensions permit. A 1×1 projection followed by global average pooling (GAP) produces the spectral context

$$\mathbf{c}_i = \text{GAP}\left(f_{\text{MobileNetV3}}(\tilde{\mathbf{S}}_i)\right) \in \mathbb{R}^{64}. \quad (5)$$

The Frontend Head (D) applies a linear classifier to $\mathbf{c}_i$, producing auxiliary emotion logits $\ell_{\text{front},i} \in \mathbb{R}^C$. The corresponding batch-mean cross-entropy is $\mathcal{L}_{\text{front}}$, labeled $\mathcal{L}_{\text{feature_enhancer}}$ in Fig. 1.

3) *Context injection: bounded node/band modulation and shift*: The spectral context determines bounded electrode- and band-specific gains:

$$\begin{aligned} [\mathbf{g}_i^{(n)}; \mathbf{g}_i^{(f)}] &= 0.25 \tanh(\mathbf{W}_g \mathbf{c}_i + \mathbf{b}_g), \\ \mathbf{g}_i^{(n)} &\in \mathbb{R}^N, \quad \mathbf{g}_i^{(f)} \in \mathbb{R}^F. \end{aligned} \quad (6)$$

The operator $[\cdot; \cdot]$ concatenates vectors, and $g_{i,n}^{(n)}$ and $g_{i,f}^{(f)}$ denote the corresponding electrode and band components. The gains are shared across time steps and applied multiplicatively:

$$X_{i,t,n,f}^{(c)} = X_{i,t,n,f} (1 + g_{i,n}^{(n)}) (1 + g_{i,f}^{(f)}). \quad (7)$$

A separate projection generates a bounded additive shift, yielding the conditioned features

$$\begin{aligned} \mathbf{D}_i &= \text{reshape}_{N \times F}(0.20 \tanh(\mathbf{W}_d \mathbf{c}_i + \mathbf{b}_d)), \\ \tilde{\mathbf{X}}_{i,t} &= \mathbf{X}_{i,t}^{(c)} + M_{i,t} \mathbf{D}_i. \end{aligned} \quad (8)$$

The shift $\mathbf{D}_i \in \mathbb{R}^{N \times F}$ is shared across time steps, and $M_{i,t}$ is broadcast over electrodes and bands. The projection parameters have dimensions $\mathbf{W}_g \in \mathbb{R}^{(N+F) \times 64}$, $\mathbf{b}_g \in \mathbb{R}^{N+F}$, $\mathbf{W}_d \in \mathbb{R}^{(NF) \times 64}$, and $\mathbf{b}_d \in \mathbb{R}^{NF}$.

4) *Temporal filtering and spatial-conditioning residuals*: For each electrode, a depthwise temporal convolution with kernel size 5 processes $\tilde{\mathbf{X}}_{i, :, n, :}$. A pointwise 1×1 convolution mixes the bands and produces $\mathbf{R}_{i, :, n, :} \in \mathbb{R}^{T \times F}$. The per-band residual coefficients are $\mathbf{w}_t = \text{sigmoid}(\boldsymbol{\theta}_t) \in (0, 1)^F$. Each component of $\boldsymbol{\theta}_t \in \mathbb{R}^F$ is initialized to $\log(0.05/0.95)$, giving an initial coefficient of 0.05. The spatial correction uses a fixed nonnegative electrode-affinity matrix $\mathbf{A} \in \mathbb{R}_{\geq 0}^{N \times N}$ shared across samples. Row normalization gives $\tilde{\mathbf{A}}_{n,m} =$

$A_{n,m} / \max(\sum_j A_{n,j}, \epsilon_A)$, with $\epsilon_A = 10^{-6}$. For the electrode vector $\tilde{\mathbf{x}}_{i,t,f} = (\tilde{X}_{i,t,1,f}, \dots, \tilde{X}_{i,t,N,f})^\top$, the graph residual is

$$\Delta \mathbf{x}_{g,i,t,f} = (\tilde{\mathbf{A}} - \mathbf{I}) \tilde{\mathbf{x}}_{i,t,f}. \quad (9)$$

Here $\mathbf{I} \in \mathbb{R}^{N \times N}$ is the identity matrix, and $\Delta X_{g,i,t,n,f}$ is the n th component of the graph residual. The per-band coefficients are $\mathbf{w}_g = 0.20 \tanh(\boldsymbol{\theta}_g) \in (-0.20, 0.20)^F$, with $\boldsymbol{\theta}_g \in \mathbb{R}^F$ initialized to zero. The signed coefficients allow either positive or negative graph corrections. The enhanced features are

$$\begin{aligned} X_{\text{enh},i,t,n,f} &= M_{i,t} (\tilde{X}_{i,t,n,f} + w_{t,f} R_{i,t,n,f} \\ &\quad + w_{g,f} \Delta X_{g,i,t,n,f}). \end{aligned} \quad (10)$$

C. Spatio-Temporal Graph Encoder

The graph attention network (GAT) encoder in Fig. 3 merges the batch and time axes of $\mathbf{X}_{\text{enh}}$ into $[BT, N, F]$ and applies two graph-attention layers independently at each time step. Let $\mathcal{G}_1$ and $\mathcal{G}_2$ denote these layers, with parameter sets Θ_1 and Θ_2 :

$$\begin{aligned} \mathbf{H}_{i,t}^{(1)} &= \mathcal{G}_1(\mathbf{X}_{\text{enh},i,t, :, :}; \Theta_1), \\ \mathbf{H}_{i,t}^{(2)} &= \mathcal{G}_2(\mathbf{H}_{i,t}^{(1)}; \Theta_2), \quad \mathbf{h}_{i,t,n} = \mathbf{H}_{i,t,n}^{(2)} \in \mathbb{R}^{32}. \end{aligned} \quad (11)$$

1) *Learned electrode-to-token pooling*: The pooling layer compresses the $N = 62$ electrode representations into $K = 16$ learned spatial tokens. Each token has a query $\mathbf{q}_k \in \mathbb{R}^{d_g}$, with $d_g = 32$. Its scaled query-electrode score is

$$e_{i,t,k,n} = \frac{\mathbf{q}_k^\top \text{LN}(\mathbf{h}_{i,t,n})}{\sqrt{d_g}}. \quad (12)$$

The scores are normalized over electrodes using softmax. The resulting weights aggregate the original electrode representations:

$$\begin{aligned} \pi_{i,t,k,n} &= \frac{\exp(e_{i,t,k,n})}{\sum_{m=1}^N \exp(e_{i,t,k,m})}, \\ \mathbf{t}_{i,t,k} &= \sum_{n=1}^N \pi_{i,t,k,n} \mathbf{h}_{i,t,n} \in \mathbb{R}^{d_g}. \end{aligned} \quad (13)$$

The K tokens are concatenated in token order:

$$\tilde{\mathbf{v}}_{i,t}^{\text{sp}} = \text{vec}(\mathbf{t}_{i,t,1}, \dots, \mathbf{t}_{i,t,K}) \in \mathbb{R}^{K d_g} = \mathbb{R}^{512}. \quad (14)$$

A layer-normalized linear projection maps the concatenated spatial representation to the 256-dimensional temporal embedding:

$$\mathbf{v}_{i,t}^{\text{sp}} = \mathbf{W}_{\text{sp}} \text{LN}(\tilde{\mathbf{v}}_{i,t}^{\text{sp}}) + \mathbf{b}_{\text{sp}} \in \mathbb{R}^{256}, \quad (15)$$

where $\mathbf{W}_{\text{sp}} \in \mathbb{R}^{256 \times 512}$ and $\mathbf{b}_{\text{sp}} \in \mathbb{R}^{256}$.

2) *Temporal modeling and shared latent embedding*: After spatial projection, a sinusoidal positional encoding $\mathbf{e}_t \in \mathbb{R}^{256}$ is added to each projected spatial feature vector $\mathbf{v}_{i,t}^{\text{sp}}$. The stacked features and positional encodings are $\mathbf{v}_{i, :}^{\text{sp}}, \mathbf{e}_{1:T} \in \mathbb{R}^{T \times 256}$. Two residual temporal-convolution blocks, with kernel sizes 3 and 5, process the sequence in that order. Each block uses a $256 \rightarrow 96 \rightarrow 96 \rightarrow 256$ bottleneck with pointwise projection, depthwise temporal convolution, batch

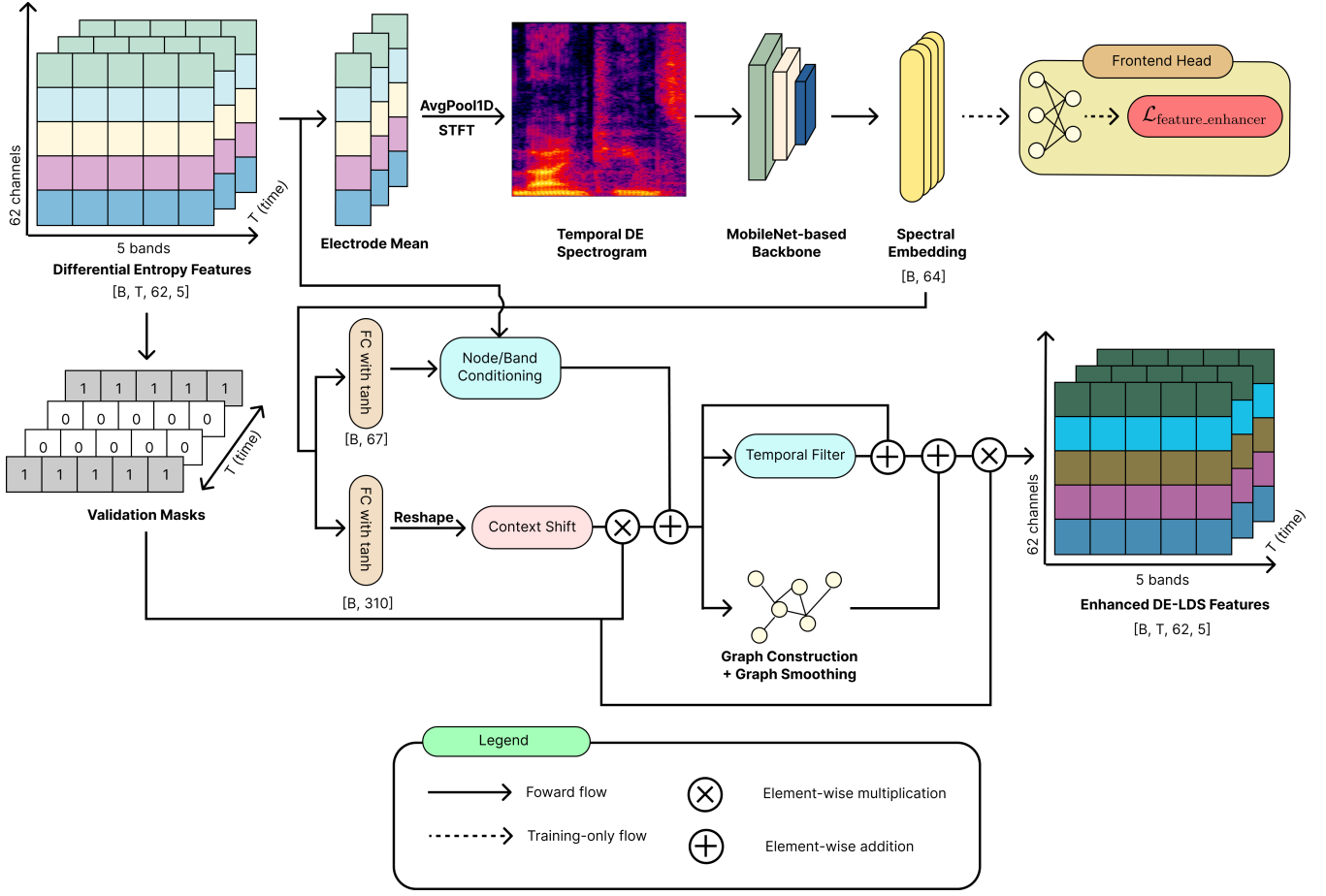


Fig. 2. Feature Enhancer. The short-time spectrum of electrode-averaged DE-LDS streams provides context for bounded electrode- and band-specific gains and an additive shift. Temporal filtering and a fixed graph produce residual corrections, which are added to the conditioned features before applying the validity mask.

normalization, GELU, dropout at rate 0.10, and residual addition. The temporal mapping is

$$\mathbf{v}_{i,:} = \mathcal{T}_5(\mathcal{T}_3(\mathbf{v}_{i,:}^{\text{sp}} + \mathbf{e}_{1:T})), \quad \mathbf{v}_{i,t} \in \mathbb{R}^{256}. \quad (16)$$

In each block, the final batch-normalization scale is initialized to zero to suppress the scale-dependent residual term at initialization. The encoded sequence is $\mathbf{v}_{i,:} \in \mathbb{R}^{T \times 256}$, and $\mathbf{v}_{i,t}$ denotes its t th row. A learned vector $\mathbf{w}_\beta \in \mathbb{R}^{256}$ and bias b_β determine the temporal attention weights:

$$\begin{aligned} a_{i,t} &= \mathbf{w}_\beta^\top \mathbf{v}_{i,t} + b_\beta, \\ \beta_{i,t} &= \frac{\exp(a_{i,t})}{\sum_{r=1}^T \exp(a_{i,r})}. \end{aligned} \quad (17)$$

The sequence is pooled as

$$\mathbf{u}_i = \sum_{t=1}^T \beta_{i,t} \mathbf{v}_{i,t} \in \mathbb{R}^{256}. \quad (18)$$

Layer normalization and a GELU-activated projection produce the shared embedding:

$$\mathbf{z}_i = \text{GELU}(\mathbf{W}_p \text{LN}(\mathbf{u}_i) + \mathbf{b}_p) \in \mathbb{R}^{192}, \quad (19)$$

with $\mathbf{W}_p \in \mathbb{R}^{192 \times 256}$ and $\mathbf{b}_p \in \mathbb{R}^{192}$.

D. Cross-Subject Representation Regularization

Heads A–C in Fig. 1 apply complementary objectives to the shared embedding $\mathbf{z}_i$. The subject-adversarial head uses gradient reversal with a logistic schedule over normalized epoch progress, slope 10, and maximum coefficient 0.10. Its linear classifier outputs subject logits $\ell_{s,i} \in \mathbb{R}^{N_s}$, supervised by batch-mean cross-entropy $\mathcal{L}_{\text{subj}}$. The contrastive head applies the outer-log supervised contrastive objective to ℓ_2 -normalized embeddings, using same-emotion positives and temperature $\tau_{\text{sc}} = 0.07$. Its loss is $\mathcal{L}_{\text{supcon}}$.

1) *Subject-adversarial objective*: The subject classifier minimizes the loss for predicting s_i , while gradient reversal sends an opposing gradient to the shared encoder. The schedule coefficient controls this reversed gradient, whereas λ_s weights the subject loss in the overall objective. The subject head is used only during training.

2) *Supervised contrastive objective*: For each anchor, positives are other minibatch samples with the same emotion label. Anchors without positives are excluded. This objective encourages class-level organization of the embeddings without explicitly using subject labels.

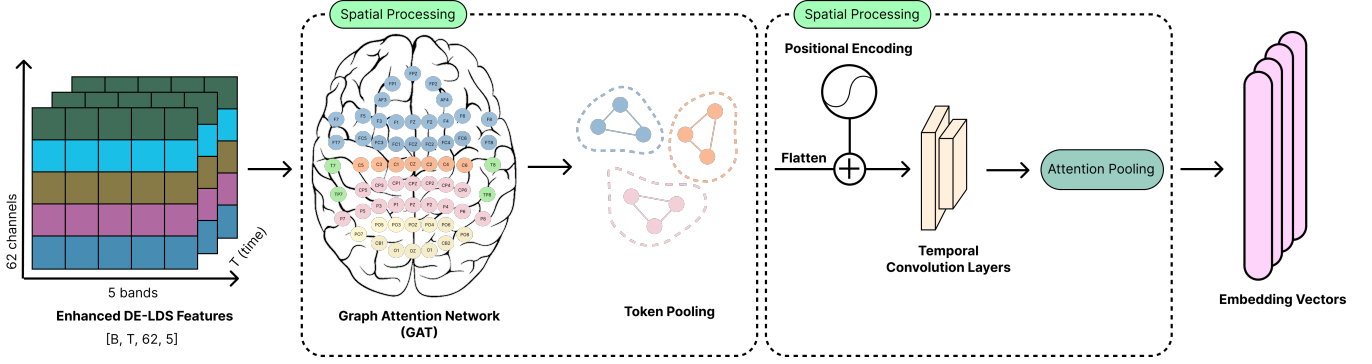


Fig. 3. Spatio-temporal graph encoder. Two graph-attention layers produce 32-dimensional electrode embeddings. Learned queries aggregate these embeddings into 16 spatial tokens, whose concatenation forms a 512-dimensional representation at each time step. Layer normalization and linear projection map this representation to 256 dimensions before positional encoding and residual temporal convolutions. Attention pooling and projection then produce the shared 192-dimensional latent representation.

3) *Cross-subject prototype alignment*: Prototypes are re-computed from subject-emotion groups in each training mini-batch. Define $\text{norm}_\epsilon(\mathbf{a}) = \mathbf{a} / \max(\|\mathbf{a}\|_2, \epsilon)$, with $\epsilon = 10^{-12}$, and $\bar{\mathbf{z}}_i = \text{norm}_\epsilon(\mathbf{z}_i) \in \mathbb{R}^{192}$. Let $\mathcal{B}_\kappa = \{i : y_i = \kappa\}$ and $\mathcal{B}_{s,\kappa} = \{i : y_i = \kappa, s_i = s\}$ denote class and subject-class index sets. Eligible classes contain at least two samples: $\mathcal{C}_B = \{\kappa \in \{1, \dots, C\} : |\mathcal{B}_\kappa| \geq 2\}$. The nonempty subject-class groups are $\Omega = \{(s, \kappa) : s \in \{1, \dots, N_s\}, \kappa \in \mathcal{C}_B, |\mathcal{B}_{s,\kappa}| > 0\}$. The global and subject-specific prototypes are

$$\begin{aligned} \mu_\kappa &= \text{norm}_\epsilon \left(\frac{1}{|\mathcal{B}_\kappa|} \sum_{i \in \mathcal{B}_\kappa} \bar{\mathbf{z}}_i \right), \quad \kappa \in \mathcal{C}_B, \\ \mu_{s,\kappa} &= \text{norm}_\epsilon \left(\frac{1}{|\mathcal{B}_{s,\kappa}|} \sum_{i \in \mathcal{B}_{s,\kappa}} \bar{\mathbf{z}}_i \right), \quad (s, \kappa) \in \Omega, \end{aligned} \quad (20)$$

Both prototypes lie in $\mathbb{R}^{192}$. The subject-to-class alignment loss is

$$\mathcal{L}_{\text{intra}} = \frac{1}{|\Omega|} \sum_{(s,\kappa) \in \Omega} (1 - \mu_{s,\kappa}^\top \mu_\kappa). \quad (21)$$

For the ordered class pairs $\mathcal{Q}_B = \{(\kappa, \kappa') : \kappa, \kappa' \in \mathcal{C}_B, \kappa \neq \kappa'\}$, the inter-class hinge loss uses margin $\delta_{\text{proto}} = 0.20$:

$$\mathcal{L}_{\text{inter}} = \frac{1}{|\mathcal{Q}_B|} \sum_{(\kappa, \kappa') \in \mathcal{Q}_B} \max(0, \mu_\kappa^\top \mu_{\kappa'} - \delta_{\text{proto}}). \quad (22)$$

The prototype objective is

$$\mathcal{L}_{\text{proto}} = \mathcal{L}_{\text{intra}} + 0.5 \mathcal{L}_{\text{inter}}. \quad (23)$$

Losses over empty index sets are set to zero: $\mathcal{L}_{\text{intra}} = 0$ if $\Omega = \emptyset$, and $\mathcal{L}_{\text{inter}} = 0$ if $\mathcal{Q}_B = \emptyset$. The latter occurs when fewer than two classes are eligible. These prototypes are minibatch statistics and are not maintained as running estimates.

E. KAN-Based Adaptive Readout

The KAN-linear operator combines a SiLU base function with a learnable univariate spline on each edge. For input $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, output coordinate o is

$$\zeta_o = \sum_{j=1}^{d_{\text{in}}} \left[W_{oj}^{(b)} \text{SiLU}(x_j) + \eta_{oj} \sum_{r=1}^8 W_{ojr}^{(s)} B_{jr}^{(3)}(x_j) \right]. \quad (24)$$

Here x_j is the j th input coordinate, $W_{oj}^{(b)}$ is a learned base weight, $W_{ojr}^{(s)}$ is a learned spline coefficient, and η_{oj} is a learned edge-specific spline scale. Here SiLU denotes the sigmoid linear unit. The function $B_{jr}^{(3)}$ is the r th cubic B-spline basis for coordinate j . The uniform grid contains five base intervals over $[-1, 1]$ and is extended by three knot intervals at each boundary, yielding eight basis functions per input coordinate. The classical classifier, quantum-angle map, measurement readout, and hybrid classifier use separate KAN-linear instances. The classical feature is $\mathbf{z}_{c,i} = \text{Dropout}_{0.20}(\mathbf{z}_i) \in \mathbb{R}^{192}$, and the corresponding logits are $\ell_{c,i} = f_{\text{KAN}}^{(c)}(\mathbf{z}_{c,i}) \in \mathbb{R}^C$. The same feature is used as the base for residual fusion. Dropout is disabled during inference.

F. Uncertainty-Gated Quantum Residual Learning

The Hybrid Classifier in Fig. 1 contains parallel classical and quantum pathways. The classical pathway produces a prediction and the entropy used for gating. The quantum pathway transforms a detached embedding into a feature residual, which is scaled and added to the classical feature before final classification.

1) *Encoding of the detached quantum input*: The quantum pathway receives a gradient-detached copy of the shared latent embedding:

$$\tilde{\mathbf{z}}_i = \text{stopgrad}(\mathbf{z}_i). \quad (25)$$

The stop-gradient operation prevents the quantum input pathway from updating the shared encoder. The encoder remains

trainable through the classical feature pathway. A two-layer projection compresses the detached embedding:

$$\begin{aligned} \mathbf{h}_{q,i}^{(1)} &= \text{GELU}(\mathbf{W}_{q,1} \text{LN}(\tilde{\mathbf{z}}_i) + \mathbf{b}_{q,1}) \in \mathbb{R}^{64}, \\ \mathbf{h}_{q,i} &= \text{GELU}(\mathbf{W}_{q,2} \mathbf{h}_{q,i}^{(1)} + \mathbf{b}_{q,2}) \in \mathbb{R}^{32}. \end{aligned} \quad (26)$$

The projection parameters have dimensions $\mathbf{W}_{q,1} \in \mathbb{R}^{64 \times 192}$, $\mathbf{b}_{q,1} \in \mathbb{R}^{64}$, $\mathbf{W}_{q,2} \in \mathbb{R}^{32 \times 64}$, and $\mathbf{b}_{q,2} \in \mathbb{R}^{32}$. A KAN-linear map followed by $\pi \tanh(\cdot)$ produces 32 bounded rotation angles:

$$\begin{aligned} \phi_i &= \text{reshape}_{4 \times 4 \times 2} \left(\pi \tanh \left(f_{\text{KAN}}^{(\phi)}(\mathbf{h}_{q,i}) \right) \right), \\ \phi_i &\in [-\pi, \pi]^{4 \times 4 \times 2}. \end{aligned} \quad (27)$$

The two angles $\phi_{i,\ell,q}^Y$ and $\phi_{i,\ell,q}^Z$ encode sample i at each of four layers and four qubits.

2) *Four-qubit data re-uploading circuit*: Let $q^+ = 1 + (q \bmod 4)$ be the next qubit on the ring. At layer $\ell \in \{1, \dots, 4\}$, each qubit $q \in \{1, \dots, 4\}$ undergoes $R_Y(\phi_{i,\ell,q}^Y)$, $R_Z(\phi_{i,\ell,q}^Z)$, and a trainable $\text{Rot}(\theta_{\ell,q,1}, \theta_{\ell,q,2}, \theta_{\ell,q,3})$ in that order. We write $\boldsymbol{\theta}_{\ell,q} = (\theta_{\ell,q,1}, \theta_{\ell,q,2}, \theta_{\ell,q,3})$ and $\boldsymbol{\Theta} \in \mathbb{R}^{4 \times 4 \times 3}$ for the full parameter tensor. Each layer then applies four ring CNOTs, with direction $q \rightarrow q^+$ on odd layers and $q^+ \rightarrow q$ on even layers. Figure 4 shows the PennyLane-generated circuit in two strips: layers 1–2 above and layers 3–4 below. The initial state is $|0\rangle^{\otimes 4}$. For final state $|\psi_i\rangle$, the measurement vector stacks the Z , X , and neighboring ZZ expectation values in that order, with increasing qubit index within each group:

$$\begin{aligned} \mathbf{m}_i &= [\{\langle \psi_i | Z_q | \psi_i \rangle\}_{q=1}^4, \{\langle \psi_i | X_q | \psi_i \rangle\}_{q=1}^4, \\ &\quad \{\langle \psi_i | Z_q Z_{q^+} | \psi_i \rangle\}_{q=1}^4] \in \mathbb{R}^{12}. \end{aligned} \quad (28)$$

A KAN-linear measurement readout maps the observables to

$$\mathbf{r}_i = f_{\text{KAN}}^{(m)}(\mathbf{m}_i) \in \mathbb{R}^{32}. \quad (29)$$

The quantum feature is projected back to the shared latent dimension:

$$\Delta \mathbf{z}_{q,i} = \mathbf{W}_\Delta \mathbf{r}_i + \mathbf{b}_\Delta \in \mathbb{R}^{192}. \quad (30)$$

The feature projection uses $\mathbf{W}_\Delta \in \mathbb{R}^{192 \times 32}$ and $\mathbf{b}_\Delta \in \mathbb{R}^{192}$. In parallel, the auxiliary Quantum Head (E) predicts a class-wise residual:

$$\hat{\boldsymbol{\delta}}_{q,i} = \mathbf{W}_q^{\text{aux}} \mathbf{r}_i + \mathbf{b}_q^{\text{aux}} \in \mathbb{R}^C. \quad (31)$$

The auxiliary parameters are $\mathbf{W}_q^{\text{aux}} \in \mathbb{R}^{C \times 32}$ and $\mathbf{b}_q^{\text{aux}} \in \mathbb{R}^C$, both initialized to zero. The loss $\mathcal{L}_{\text{quantum}}$ trains this head to predict a signed classwise residual from $\mathbf{r}_i$. This C -dimensional output is distinct from the 192-dimensional feature residual $\Delta \mathbf{z}_{q,i}$ used for fusion.

3) *Uncertainty-gated residual fusion*: The gate is computed from the classical logits $\ell_{c,i}$ using detached class probabilities:

$$p_{i,k}^{(c)} = \text{stopgrad} \left(\frac{\exp(\ell_{c,i,k})}{\sum_{j=1}^C \exp(\ell_{c,i,j})} \right). \quad (32)$$

Here $\ell_{c,i,k}$ is the k th component of $\ell_{c,i}$, and $\mathbf{p}_i^{(c)} = (p_{i,1}^{(c)}, \dots, p_{i,C}^{(c)})^\top$. A numerically stabilized approximation to normalized predictive entropy is

$$\bar{\mathcal{H}}_i = -\frac{1}{\log C} \sum_{k=1}^C p_{i,k}^{(c)} \log(p_{i,k}^{(c)} + \epsilon_p), \quad \epsilon_p = 10^{-8}. \quad (33)$$

The gate contains no learned parameters and is defined as

$$g_{u,i} = [\text{clip}_{[0,1]}(\bar{\mathcal{H}}_i)]^\gamma, \quad \gamma = 1.5. \quad (34)$$

Entropy is used as a deterministic gating signal, not as a calibrated estimate of prediction error. Let $E \geq 1$ be the number of training epochs, $e \in \{0, \dots, E-1\}$ the zero-based epoch index, and $\rho_e = e / \max(E-1, 1)$ the normalized progress. A learned scalar s_q , initialized to zero, determines the fusion coefficient together with a linear warm-up:

$$\begin{aligned} \alpha_q &= \alpha_{\max} \text{sigmoid}(s_q) r_q(\rho_e), \\ r_q(\rho) &= \text{clip}_{[0,1]} \left(\frac{\rho - 0.10}{0.20} \right), \quad 0 \leq \rho \leq 1. \end{aligned} \quad (35)$$

With $\alpha_{\max} = 0.25$, the warm-up factor is zero during the first 10% of training, increases linearly until 30%, and then remains at one. The coefficient satisfies $0 \leq \alpha_q \leq 0.25$ and equals $0.125 r_q(\rho_e)$ when $s_q = 0$. This constrains the scalar coefficient rather than the magnitude of the feature residual. Fusion is performed as

$$\mathbf{z}_{h,i} = \mathbf{z}_{c,i} + g_{u,i} \alpha_q \Delta \mathbf{z}_{q,i}. \quad (36)$$

The final emotion logits are

$$\ell_i = f_{\text{KAN}}^{(h)}(\mathbf{z}_{h,i}) \in \mathbb{R}^C. \quad (37)$$

The hybrid and classical KAN classifiers have independent parameters. Circuit evaluation uses a state-vector simulator; no quantum computational advantage is assumed.

G. Optimization Objective

The fused emotion, classical, frontend, and subject losses, denoted $\mathcal{L}_{\text{emo}}$, $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{front}}$, and $\mathcal{L}_{\text{subj}}$, are batch-mean cross-entropies applied to ℓ_i , $\ell_{c,i}$, $\ell_{\text{front},i}$, and $\ell_{s,i}$, respectively. The contrastive and prototype losses are $\mathcal{L}_{\text{supcon}}$ and $\mathcal{L}_{\text{proto}}$, as specified above. For one-hot emotion label $\mathbf{e}_{y_i} \in \mathbb{R}^C$ and detached classical probability vector $\mathbf{p}_i^{(c)}$, the auxiliary quantum target is

$$\boldsymbol{\delta}_{q,i} = \mathbf{e}_{y_i} - \mathbf{p}_i^{(c)}. \quad (38)$$

Its auxiliary objective is

$$\mathcal{L}_{\text{quantum}} = \frac{1}{BC} \sum_{i=1}^B \left\| \hat{\boldsymbol{\delta}}_{q,i} - \boldsymbol{\delta}_{q,i} \right\|_2^2. \quad (39)$$

The target is recomputed from the current classical prediction at each forward pass, with stop-gradient preventing backpropagation through the probabilities. For nonnegative weights $\lambda_c, \lambda_f, \lambda_s, \lambda_{sc}, \lambda_p, \lambda_q$, the overall training objective is

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{emo}} + \lambda_c \mathcal{L}_{\text{cls}} + \lambda_f \mathcal{L}_{\text{front}} \\ &\quad + \lambda_s \mathcal{L}_{\text{subj}} + \lambda_{sc} \mathcal{L}_{\text{supcon}} \\ &\quad + \lambda_p \mathcal{L}_{\text{proto}} + \lambda_q \mathcal{L}_{\text{quantum}}. \end{aligned} \quad (40)$$

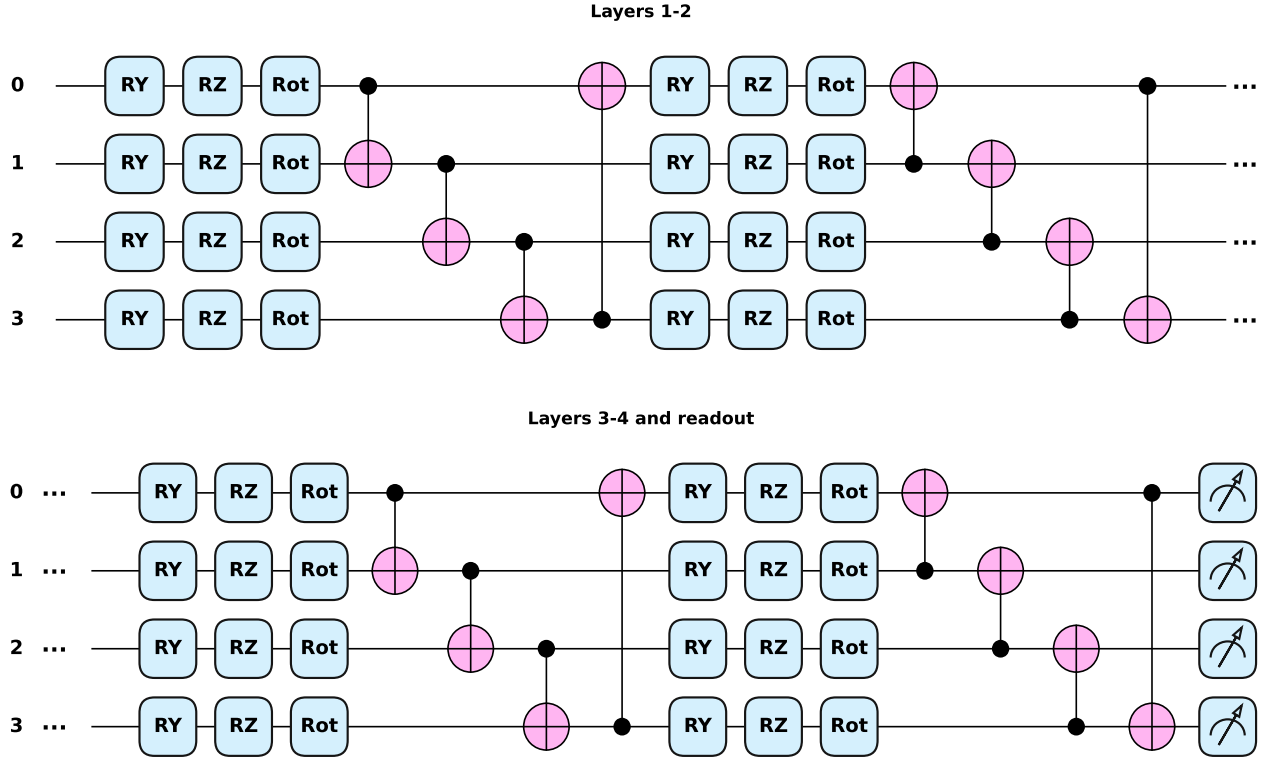


Fig. 4. Four-qubit, four-layer PQC in the quantum branch. Each layer applies data-dependent R_Y and R_Z rotations, trainable $\text{Rot}(\theta_{\ell,q})$ gates, and four ring CNOTs with alternating direction. The two strips show layers 1–2 and 3–4. The readout comprises four Z_q , four X_q , and four neighboring $Z_q Z_{q+}$ expectation values. A KAN-linear map converts these 12 values into a 32-dimensional representation for the auxiliary residual-prediction head and the 32-to-192 feature projection.

The quantum loss supervises the auxiliary residual-prediction head and its input representation. The fused emotion loss also backpropagates through the quantum feature residual when the gated fusion pathway is active. The subject head minimizes cross-entropy, while gradient reversal changes the sign of the gradient passed to the shared encoder. Figure 1 shows the corresponding training branches.

IV. EXPERIMENTAL SETUP

Experiments use a strict leave-one-subject-out (LOSO) protocol. For a dataset containing S subjects, each fold holds out one subject for final testing and trains exclusively on the other $S - 1$ source subjects. The held-out subject supplies neither observations nor labels for training, hyperparameter tuning, or checkpoint selection and is evaluated only after training is complete. Any validation or model-selection step is restricted to source subjects. Repeating this procedure over all subjects gives 15 folds for SEED, 15 for SEED IV, 16 for SEED V, and 20 for SEED VII. Thus, the methodology’s N_s denotes the $S - 1$ training subjects within a fold.

The model operates on precomputed five-band DE-LDS features from 62 EEG channels using all available sessions. For each configuration and held-out subject, training and final evaluation are repeated with five random seeds. Accuracy

and macro-averaged class F1 are first averaged across the five seeds within each subject fold. The reported mean $\pm$ standard deviation (SD) is then computed across these subject-level averages, with equal weight assigned to each held-out subject. All ablations retain the same subject partitions and preprocessing as the complete model. Thus, SD describes between-subject variation after seed averaging; it is neither a confidence interval nor an estimate of variability across seeds. Table I summarizes the implementation and training settings.

For computational profiling, the standardized input has shape $[1, 32, 62, 5]$. We distinguish stored model parameters, pure 32-bit floating-point (FP32) weight storage, profiler-counted multiply-accumulate operations (MACs), and floating-point operations (FLOPs). The profiler uses two FLOPs per counted MAC; both quantities are rounded independently for reporting. Its counts exclude operations that are not fully covered, including the STFT, attention and graph tensor contractions, softmax, normalization, elementwise operations, and quantum state-vector gates. They therefore provide partial operation counts rather than end-to-end inference cost. Quantum simulation estimates are reported separately.

TABLE I
IMPLEMENTATION AND TRAINING SETTINGS.

Setting	Configuration
EEG channels	62
Frequency bands	5
Spatial GAT layers	2
Spatial tokens	16
Graph hidden dimension	32
Embedding dimension	256
Temporal bottleneck	96
Temporal kernels	3, 5
Qubits	4
Quantum layers	4
Quantum dimension	32
KAN grid size	5
KAN spline order	3
Entropy exponent γ	1.5
Batch size	32
Optimizer	AdamW
Backbone LR	3×10^{-4}
Head/quantum LR	5×10^{-4}
Front-end LR	8×10^{-4}
Weight decay	10^{-4}
LR scheduler	Cosine annealing
Minimum LR	10^{-6}
Maximum epochs	300
Random seeds	5
Gaussian noise	0.035
Gain jitter	0.10
Channel drop	0.05
Time mask	0.10
Band drop	0.15

V. EXPERIMENTAL RESULTS

A. Comparison with Reported Cross-Subject Results

Table II summarizes the reported cross-subject results on SEED, SEED IV, SEED V, and SEED VII. QuanKAN achieves accuracies of $87.41 \pm 6.05\%$, $78.43 \pm 4.48\%$, $86.81 \pm 3.72\%$, and $73.81 \pm 4.58\%$, respectively. It attains the highest listed accuracy on SEED V and SEED VII, exceeding the strongest reported comparators, Chen *et al.* [38] and Yang *et al.* [39], by 9.69 and 14.58 percentage points. On SEED IV, QuanKAN ranks second among the listed methods, 1.04 percentage points below Wang *et al.* [40]. On SEED, its accuracy is 3.26 percentage points below the highest listed value, reported by Rao *et al.* [41]. QuanKAN also obtains the highest listed F1 score on each dataset, with values of $87.12 \pm 6.64\%$, $78.12 \pm 4.67\%$, $86.60 \pm 3.95\%$, and $73.16 \pm 4.91\%$, respectively. These exceed the highest available comparator F1 scores by 1.92, 8.07, 19.55, and 17.34 percentage points. This comparison is restricted to methods that report F1 scores, which are less widely available than accuracy. The table shows the largest accuracy margins on SEED V and SEED VII, alongside competitive results on SEED and SEED IV. Because these values come from separate studies, the differences are descriptive and do not establish statistical significance or isolate the contribution of any individual QuanKAN component.

B. Ablation Studies

We further evaluate the architectural and optimization choices of QuanKAN through six groups of ablation exper-

iments covering the major model components, the spatio-temporal backbone, quantum-circuit design, quantum-classical interface, auxiliary objectives, and alternative prediction heads. Full QuanKAN provides the common reference, and each variant changes the named architectural or objective choice. Component suppression can leave a module and its stored weights instantiated; consequently, disabling a pathway need not reduce the parameter count. Tables III–VIII report the predictive results under strict LOSO evaluation.

1) *Contribution of Major Components*: Table III shows the largest reductions when the front-end or spectral conditioning is disabled. Without spectral conditioning, accuracy decreases by 16.45, 21.86, 34.73, and 24.81 percentage points on SEED, SEED IV, SEED V, and SEED VII, respectively. Disabling the entire front-end yields a similar pattern. Graph diffusion, temporal residual filtering, and the quantum branch also contribute: removing the quantum branch reduces accuracy by 6.82, 10.93, 17.78, and 12.75 percentage points. These comparisons identify input refinement as the most influential component group among the tested removals, while also favoring the complete downstream architecture.

2) *Spatio-Temporal Backbone Design*: Table IV evaluates graph depth, positional encoding, and spatial and temporal aggregation. Removing the second GAT layer reduces accuracy to 80.00%, 68.52%, 71.25%, and 61.87% across the four datasets. Positional encoding is particularly consequential on SEED V, where its removal lowers accuracy to 63.47%. Mean spatial or temporal pooling also remains below the complete model. The relative ordering of these alternatives varies across datasets, indicating that the effects of graph depth and learned aggregation depend on the dataset.

3) *Quantum Circuit and Measurement Design*: Table V compares removal of entanglement, removal of trainable rotations, and restriction to Z measurements. Each variant has lower accuracy and F1 than Full QuanKAN on all four datasets. The no-rotation variant retains the highest ablated accuracy on SEED, whereas the Z -only variant is strongest on the other datasets. The complete circuit therefore provides the most effective tested combination of encoding, trainable transformations, and observables. These removals evaluate architectural contributions, but do not independently identify the physical source of a quantum advantage.

4) *Quantum Gating and KAN Interface*: Table VI evaluates entropy gating, learned residual scaling, and the KAN mappings surrounding the circuit. Removing the entropy gate decreases accuracy by 7.56, 11.12, 17.92, and 11.69 percentage points. Fixing the quantum scale or replacing either KAN map with a linear transformation also lowers performance. The results favor the complete interface, with its adaptive input and readout mappings and sample-dependent gate, over each tested simplification.

5) *Auxiliary Training Objectives*: Table VII evaluates subject-adversarial, supervised contrastive, prototype, and quantum auxiliary objectives. Every single-loss removal reduces accuracy and F1 relative to the complete objective. Removing supervised contrastive learning produces the smallest

TABLE II

REPORTED CROSS-SUBJECT EEG EMOTION RECOGNITION RESULTS ON FOUR SEED DATASETS. ACCURACY AND F1 SCORES ARE EXPRESSED AS PERCENTAGES; $\pm$ VALUES REPRODUCE THE VARIABILITY REPORTED BY THE RESPECTIVE STUDIES, AND – DENOTES AN UNREPORTED METRIC. FOR QUANKAN, METRICS ARE AVERAGED OVER FIVE SEEDS WITHIN EACH SUBJECT FOLD BEFORE COMPUTING MEAN $\pm$ SD ACROSS SUBJECTS. RESULTS FROM SEPARATE STUDIES ARE DESCRIPTIVE COMPARISONS.

Studies	Year	SEED		SEED IV		SEED V		SEED VII	
		Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Zhou <i>et al.</i> [42]	2023	–	–	76.96 $\pm$ 7.19	–	71.40 $\pm$ 9.43	–	–	–
She <i>et al.</i> [43]	2023	86.16 $\pm$ 7.87	–	59.29 $\pm$ 13.65	–	65.59 $\pm$ 4.41	–	–	–
Zhang <i>et al.</i> [44]	2023	88.75 $\pm$ 8.68	–	74.49 $\pm$ 13.80	–	–	–	–	–
Pan <i>et al.</i> [45]	2023	85.90 $\pm$ 4.90	–	76.37 $\pm$ 5.77	–	–	–	–	–
Jiménez-Guarneros <i>et al.</i> [46]	2023	85.27	85.20	69.58	68.90	–	–	–	–
Aslan <i>et al.</i> [47]	2024	82.06	–	–	–	–	–	–	–
Chen <i>et al.</i> [38]	2024	–	–	77.16 $\pm$ 4.73	–	77.12 $\pm$ 10.87	–	–	–
Zhang <i>et al.</i> [48]	2024	87.51 $\pm$ 7.68	–	67.80 $\pm$ 8.25	–	–	–	–	–
Li <i>et al.</i> [49]	2025	75.71	71.62	61.64	58.48	54.47	50.71	–	–
Wang <i>et al.</i> [40]	2025	87.02 $\pm$ 4.36	–	79.47 $\pm$ 25.89	–	–	–	–	–
Tan <i>et al.</i> [50]	2025	86.39 $\pm$ 12.51	–	77.80 $\pm$ 6.49	–	–	–	–	–
Zhu <i>et al.</i> [51]	2025	87.30 $\pm$ 7.89	–	72.61 $\pm$ 11.70	–	–	–	–	–
Jimenez-Guarneros <i>et al.</i> [52]	2025	69.44	67.25	71.00	70.05	71.90	67.05	–	–
Zhang <i>et al.</i> [31]	2025	86.37 $\pm$ 8.13	–	69.38 $\pm$ 8.77	–	–	–	–	–
Zhong <i>et al.</i> [53]	2025	89.44 $\pm$ 4.89	–	74.57 $\pm$ 6.18	–	69.15 $\pm$ 9.76	–	–	–
Wang <i>et al.</i> [54]	2025	85.68 $\pm$ 4.16	–	73.94 $\pm$ 10.42	–	71.22 $\pm$ 16.21	–	51.12 $\pm$ 8.51	–
Liu <i>et al.</i> [55]	2025	87.32	–	73.42	–	–	–	–	–
Wu <i>et al.</i> [56]	2026	77.45 $\pm$ 11.28	–	53.04 $\pm$ 12.95	–	51.98 $\pm$ 10.57	–	33.05 $\pm$ 5.07	–
Rao <i>et al.</i> [41]	2026	90.67 $\pm$ 4.09	–	62.60 $\pm$ 4.32	–	–	–	–	–
Li <i>et al.</i> [57]	2026	79.51 $\pm$ 6.09	68.12 $\pm$ 13.24	–	–	–	–	–	–
Li <i>et al.</i> [58]	2026	79.34 $\pm$ 6.34	76.72	63.16 $\pm$ 9.03	60.39	57.53 $\pm$ 9.72	56.27	–	–
Yang <i>et al.</i> [39]	2026	85.65 $\pm$ 6.87	–	75.55 $\pm$ 7.94	–	–	–	59.23 $\pm$ 10.87	55.82 $\pm$ 12.08
Wang <i>et al.</i> [59]	2026	–	–	–	–	–	–	49.60 $\pm$ 8.60	49.50 $\pm$ 8.60
Wang <i>et al.</i> [60]	2026	77.90 $\pm$ 13.17	75.64 $\pm$ 15.18	68.01 $\pm$ 12.99	66.70 $\pm$ 13.23	60.80 $\pm$ 16.40	59.63 $\pm$ 16.31	46.29 $\pm$ 12.86	42.74 $\pm$ 14.01
QuanKAN (Ours)	2026	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91

TABLE III

ABLATION OF THE MAJOR QUANKAN COMPONENTS. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
w/o Frontend	72.89 $\pm$ 11.54	70.58 $\pm$ 13.69	56.67 $\pm$ 5.25	54.93 $\pm$ 5.76	54.17 $\pm$ 5.93	52.83 $\pm$ 6.39	48.94 $\pm$ 5.34	47.74 $\pm$ 5.92
w/o Graph Diffusion	79.26 $\pm$ 8.06	78.78 $\pm$ 8.99	67.41 $\pm$ 4.12	66.57 $\pm$ 4.97	70.56 $\pm$ 11.48	69.27 $\pm$ 11.90	61.94 $\pm$ 5.92	60.12 $\pm$ 6.83
w/o Quantum Branch	80.59 $\pm$ 7.89	79.94 $\pm$ 8.73	67.50 $\pm$ 3.82	66.82 $\pm$ 4.41	69.03 $\pm$ 11.95	67.24 $\pm$ 13.35	61.06 $\pm$ 4.89	59.27 $\pm$ 6.64
w/o Spectral Conditioning	70.96 $\pm$ 11.08	68.46 $\pm$ 13.46	56.57 $\pm$ 5.08	55.10 $\pm$ 5.88	52.08 $\pm$ 5.27	50.11 $\pm$ 6.05	49.00 $\pm$ 5.28	47.48 $\pm$ 6.37
w/o Temporal Residual	80.00 $\pm$ 6.13	79.55 $\pm$ 6.57	67.22 $\pm$ 3.79	66.73 $\pm$ 4.24	68.75 $\pm$ 10.50	66.88 $\pm$ 11.68	62.56 $\pm$ 3.27	61.19 $\pm$ 4.63

TABLE IV

ABLATION OF THE SPATIO-TEMPORAL BACKBONE. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
w/o Second Spatial GAT	80.00 $\pm$ 8.31	79.12 $\pm$ 9.45	68.52 $\pm$ 5.14	67.94 $\pm$ 5.67	71.25 $\pm$ 9.67	69.11 $\pm$ 11.28	61.87 $\pm$ 7.10	59.52 $\pm$ 8.95
w/o Positional Encoding	80.89 $\pm$ 5.37	80.35 $\pm$ 6.02	68.43 $\pm$ 5.75	67.48 $\pm$ 6.77	63.47 $\pm$ 11.60	61.07 $\pm$ 12.62	61.00 $\pm$ 6.50	58.86 $\pm$ 8.15
Mean Spatial Pooling	80.44 $\pm$ 8.52	79.86 $\pm$ 9.43	67.59 $\pm$ 4.96	66.73 $\pm$ 5.72	69.31 $\pm$ 9.66	67.83 $\pm$ 10.99	61.50 $\pm$ 4.67	59.86 $\pm$ 5.79
Mean Temporal Pooling	78.37 $\pm$ 7.14	77.49 $\pm$ 8.26	68.06 $\pm$ 5.37	67.36 $\pm$ 5.55	66.94 $\pm$ 9.19	64.23 $\pm$ 10.92	61.38 $\pm$ 6.55	60.14 $\pm$ 7.56

accuracy reduction among these variants on SEED IV, SEED V, and SEED VII, whereas removing the subject-adversarial objective retains the highest ablated accuracy on SEED. The ordering shows that the influence of each training signal varies by dataset; it does not support assigning one universal importance ranking to the objectives.

6) *Alternative Quantum and Hybrid Heads:* Table VIII compares the hybrid design with a matched multilayer perceptron (MLP) final head, a capacity-matched classical residual, and a quantum-only head. The MLP final head yields accuracies of 79.70%, 66.20%, 63.89%, and 59.00%. The capacity-

matched classical residual also remains below Full QuanKAN despite nearly identical stored parameter counts, as detailed in the model-size analysis below. The quantum-only head produces the lowest accuracy among these alternatives on SEED IV, SEED V, and SEED VII. Together, these controls favor the tested quantum residual configuration within the hybrid model; parameter matching alone does not establish equivalence of representational capacity or optimization behavior.

TABLE V

ABLATION OF THE QUANTUM CIRCUIT AND MEASUREMENT DESIGN. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
w/o Quantum Entanglement	78.67 $\pm$ 8.50	77.16 $\pm$ 10.06	67.50 $\pm$ 4.33	66.74 $\pm$ 4.95	69.31 $\pm$ 11.71	67.79 $\pm$ 12.81	61.88 $\pm$ 4.34	60.12 $\pm$ 5.61
w/o Trainable Rotations	81.33 $\pm$ 6.73	80.71 $\pm$ 7.49	67.50 $\pm$ 4.56	66.85 $\pm$ 5.30	68.75 $\pm$ 10.29	67.29 $\pm$ 11.20	61.12 $\pm$ 5.52	59.26 $\pm$ 6.50
Z-only Measurements	80.15 $\pm$ 7.28	79.66 $\pm$ 7.94	68.43 $\pm$ 4.49	67.81 $\pm$ 5.05	69.58 $\pm$ 11.10	67.95 $\pm$ 12.96	62.44 $\pm$ 5.46	61.00 $\pm$ 5.67

TABLE VI

ABLATION OF QUANTUM GATING AND THE KAN INTERFACE. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
w/o Entropy Gate	79.85 $\pm$ 8.01	79.40 $\pm$ 8.83	67.31 $\pm$ 4.42	66.56 $\pm$ 5.10	68.89 $\pm$ 10.89	67.34 $\pm$ 12.15	62.12 $\pm$ 4.68	60.58 $\pm$ 5.71
Fixed Quantum Scale	80.30 $\pm$ 8.11	79.58 $\pm$ 9.27	66.39 $\pm$ 4.43	65.73 $\pm$ 5.00	69.58 $\pm$ 11.05	68.28 $\pm$ 11.51	61.75 $\pm$ 5.07	59.95 $\pm$ 6.49
Linear Quantum-Input Map	80.74 $\pm$ 6.67	79.83 $\pm$ 7.99	68.89 $\pm$ 4.70	68.17 $\pm$ 4.93	67.64 $\pm$ 8.16	65.49 $\pm$ 9.63	62.31 $\pm$ 3.52	60.72 $\pm$ 4.18
Linear Quantum-Output Map	79.41 $\pm$ 5.10	78.96 $\pm$ 5.37	68.24 $\pm$ 3.40	67.59 $\pm$ 3.79	70.14 $\pm$ 7.07	68.88 $\pm$ 7.70	63.63 $\pm$ 5.60	62.15 $\pm$ 6.76

TABLE VII

ABLATION OF AUXILIARY TRAINING OBJECTIVES. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
w/o SupCon Loss	79.11 $\pm$ 9.63	78.11 $\pm$ 11.36	68.61 $\pm$ 5.57	67.42 $\pm$ 6.66	70.69 $\pm$ 10.01	69.22 $\pm$ 11.15	65.56 $\pm$ 6.23	64.08 $\pm$ 7.86
w/o Prototype Loss	80.59 $\pm$ 7.28	79.87 $\pm$ 8.30	67.87 $\pm$ 4.86	67.36 $\pm$ 5.22	66.94 $\pm$ 12.67	65.07 $\pm$ 14.70	62.00 $\pm$ 6.04	60.27 $\pm$ 7.15
w/o Subject Adversary	81.48 $\pm$ 7.24	80.78 $\pm$ 8.03	67.31 $\pm$ 4.73	66.56 $\pm$ 5.29	69.58 $\pm$ 9.02	68.26 $\pm$ 9.61	61.19 $\pm$ 4.63	59.55 $\pm$ 5.83
w/o Quantum Auxiliary Loss	81.19 $\pm$ 8.54	80.64 $\pm$ 9.30	67.31 $\pm$ 4.33	66.59 $\pm$ 4.83	70.00 $\pm$ 9.97	68.77 $\pm$ 10.81	61.50 $\pm$ 5.67	59.98 $\pm$ 6.70

TABLE VIII

ALTERNATIVE QUANTUM AND HYBRID CLASSIFICATION HEADS. ACCURACY AND MACRO-F1 (%) ARE AVERAGED OVER FIVE SEEDS WITHIN EACH HELD-OUT-SUBJECT FOLD, THEN REPORTED AS MEAN $\pm$ SD ACROSS SUBJECTS UNDER STRICT LOSO EVALUATION.

Variant	SEED		SEED IV		SEED V		SEED VII	
	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)	Accuracy (%)	F1 Score (%)
Full QuanKAN	87.41 $\pm$ 6.05	87.12 $\pm$ 6.64	78.43 $\pm$ 4.48	78.12 $\pm$ 4.67	86.81 $\pm$ 3.72	86.60 $\pm$ 3.95	73.81 $\pm$ 4.58	73.16 $\pm$ 4.91
Matched MLP Final Head	79.70 $\pm$ 9.42	78.83 $\pm$ 10.59	66.20 $\pm$ 5.64	65.54 $\pm$ 5.88	63.89 $\pm$ 9.28	62.34 $\pm$ 10.36	59.00 $\pm$ 6.67	57.08 $\pm$ 7.60
Capacity-Matched Classical Residual	78.52 $\pm$ 9.86	77.37 $\pm$ 11.71	68.06 $\pm$ 5.34	67.27 $\pm$ 5.65	69.72 $\pm$ 9.39	68.39 $\pm$ 10.40	63.56 $\pm$ 6.70	61.69 $\pm$ 8.34
Quantum-Only Head	78.52 $\pm$ 7.33	77.99 $\pm$ 7.81	62.41 $\pm$ 5.21	61.35 $\pm$ 5.70	56.25 $\pm$ 8.12	51.17 $\pm$ 10.37	54.19 $\pm$ 6.04	51.13 $\pm$ 7.64

C. Classwise Performance

Table IX presents classwise metrics computed from the pooled held-out confusion matrices. They complement the subject-averaged accuracy and F1 results without replacing their aggregation. SEED has closely grouped class F1 scores, while SEED IV has its lowest class F1 for Fear (0.754). SEED V is reported using the class indices used in the confusion-matrix analysis. On SEED VII, Sad has the lowest F1 (0.661), followed by Neutral (0.680); Surprise and Disgust reach 0.809 and 0.801, respectively. These results reveal class-dependent difficulty that remains despite the stronger aggregate performance of the full architecture.

D. Predictive-Entropy Gate and Residual Behavior

Table X summarizes one post-training diagnostic pass over 4,075 held-out samples from the Full QuanKAN checkpoints. Samples are grouped by the correctness of the classical branch, rather than the final hybrid prediction. The mean gate is higher

for incorrect classical predictions on every dataset, consistent with the gate’s dependence on predictive entropy. This association is descriptive and does not establish calibration or show that the residual corrects each classical error.

The effective residual norm is higher in the incorrect group on SEED IV, SEED V, and SEED VII, but slightly lower on SEED despite its higher gate. This distinction follows from the fusion mechanism: the correction depends jointly on the gate, fusion coefficient, and unscaled feature residual. Its mean norm is small relative to the mean classical feature norm in every group. These are aggregate norm comparisons, not per-sample bounds or guarantees of prediction stability.

E. Exploratory Subject-Paired Analysis

Table XI reports the paired analyses for spectral-conditioning removal, quantum-branch removal, and the quantum-only head. Subject folds are matched between Full QuanKAN and each variant. Paired t tests and Wilcoxon

TABLE IX
CLASSWISE PRECISION, RECALL, AND F1 FOR THE FULL QUANAKAN
CONFUSION-MATRIX ANALYSIS. METRICS LIE IN $[0, 1]$ AND ARE
COMPUTED FROM POOLED HELD-OUT PREDICTIONS; SUPPORT IS THE
NUMBER OF LABELED SAMPLES. THESE POOLED METRICS ARE DISTINCT
FROM THE SUBJECT-AVERAGED RESULTS IN TABLE II. SEED V LABELS
RETAIN THE CLASS INDICES USED IN THE ANALYSIS.

Dataset	Class	Precision	Recall	F1	Support
SEED	Negative	0.871	0.871	0.871	225
SEED	Neutral	0.873	0.884	0.879	225
SEED	Positive	0.878	0.867	0.872	225
SEED IV	Neutral	0.747	0.778	0.762	270
SEED IV	Sad	0.834	0.781	0.807	270
SEED IV	Fear	0.771	0.737	0.754	270
SEED IV	Happy	0.788	0.841	0.814	270
SEED V	Class0	0.979	0.972	0.976	144
SEED V	Class1	0.945	0.840	0.890	144
SEED V	Class2	0.833	0.833	0.833	144
SEED V	Class3	0.771	0.840	0.804	144
SEED V	Class4	0.831	0.854	0.842	144
SEED VII	Anger	0.742	0.721	0.732	240
SEED VII	Disgust	0.815	0.787	0.801	240
SEED VII	Fear	0.728	0.725	0.727	240
SEED VII	Happy	0.754	0.729	0.742	240
SEED VII	Neutral	0.609	0.769	0.680	160
SEED VII	Sad	0.691	0.633	0.661	240
SEED VII	Surprise	0.806	0.812	0.809	240

signed-rank tests use Holm adjustment over 12 comparisons within each test family. Displayed mean differences use the rounded values in the main accuracy tables, whereas p-values and paired Cohen’s effect sizes d_z are based on subject-paired outcomes.

All adjusted p-values are below 0.001, and the paired effect sizes are positive. They provide exploratory evidence for consistent subject-level differences in the analyzed runs. However, LOSO folds have overlapping source-training sets, so fold outcomes are not independent experimental replications. The tests compare subject-level scores after averaging over five seeds within each fold; they do not separately quantify variability across seeds. Accordingly, these tests complement the ablation results and are not used to establish confirmatory statistical superiority over external methods.

F. Model Size and Computational Cost

Table XII reports the model’s stored parameter budget and weight footprint. Full QuanKAN contains 589,366 parameters on SEED and 691,652 on SEED VII, corresponding to 2.25–2.64 MiB of pure FP32 weights. This footprint excludes optimizer state, activations, buffers, and file-format overhead. The standardized forward pass has 20.10–20.12 million counted MACs, or 40.19–40.24 million counted FLOPs. Because the profiler omits several operation families, these values characterize the covered classical operations rather than total inference complexity.

Functional component suppression and stored parameter removal have different effects on model size. The major component-removal variants retain the full reference parameter

count. Removing the second spatial GAT reduces the SEED count from 20.10 to 11.72 million MACs while reducing stored parameters by only 4,160. This illustrates that parameter count and sequence-processing cost capture different aspects of the architecture. On SEED, replacing the final head with the matched MLP yields 589,489 parameters, and the capacity-matched classical residual yields 589,399, compared with 589,366 for Full QuanKAN. Their lower predictive scores therefore occur with nearly unchanged stored model size.

Table XIII reports the separate state-vector arithmetic estimates. The four-qubit, four-layer reference uses 16 amplitudes, 48 logical one-qubit gate applications, and an estimated 11,300 gate/readout FLOPs per sample. Increasing to eight qubits expands the state to 256 amplitudes and the estimate to 361,192 FLOPs, whereas the classical profiler changes only slightly. These estimates demonstrate why quantum simulation cost must be examined separately. They do not measure wall-clock execution or support a claim of reduced physical-hardware cost.

VI. DISCUSSION

A. Input Conditioning and Cross-Subject Representation Learning

The largest component-removal effects occur before graph and temporal encoding. This pattern supports the use of DE-stream context to guide input refinement: electrode–band gains and context shifts alter the features presented to the encoder, while temporal and graph residuals provide further corrections. The context is derived from the variation of precomputed DE-LDS trajectories, so the mechanism should be interpreted as feature conditioning rather than recovery of new raw-EEG spectral information.

The backbone and objective ablations favor combining spatial relationships, temporal organization, and source-subject regularization. Subject-adversarial, contrastive, and prototype losses constrain different aspects of the embedding, but lower subject predictability and stronger class alignment are training objectives rather than guarantees of invariance. Held-out-subject performance provides the empirical evidence for generalization. The classwise analysis also reveals remaining unevenness, especially the lower Sad and Neutral F1 scores on SEED VII, which aggregate scores alone do not expose.

B. What the Quantum Residual Evidence Supports

The alternative-head and interface experiments support the quantum branch’s assigned role as a selectively weighted feature correction. The weaker quantum-only head shows that its usefulness depends on interaction with the classical pathway. The capacity-matched classical residual provides a relevant control against an explanation based solely on adding stored parameters, although the compared mappings can still differ in expressivity and trainability. Likewise, single-component circuit ablations establish sensitivity to the chosen design without identifying entanglement as the unique cause of the performance differences.

TABLE X

POST-TRAINING DIAGNOSTICS GROUPED BY CORRECTNESS OF THE CLASSICAL PREDICTION ON 4,075 HELD-OUT SAMPLES. GATE AND NORM ENTRIES ARE SAMPLE-LEVEL MEAN $\pm$ SD, DISTINCT FROM THE FOLD-LEVEL SD IN THE ACCURACY/F1 TABLES. THE LAST COLUMN IS THE NORM OF THE SCALED FEATURE CORRECTION.

Dataset	Classical prediction	Samples	g_u	$\ \mathbf{z}_c\ _2$	$\ g_u \alpha_q \Delta \mathbf{z}_q\ _2$
SEED	Correct	586	0.2520 ± 0.2215	24.9254 ± 3.9909	0.1901 ± 0.4926
SEED	Incorrect	89	0.3189 ± 0.2323	24.0887 ± 3.2902	0.1760 ± 0.3795
SEED IV	Correct	841	0.2829 ± 0.2069	23.4164 ± 2.1389	0.1633 ± 0.2714
SEED IV	Incorrect	239	0.3825 ± 0.1976	23.3571 ± 2.2623	0.1926 ± 0.2876
SEED V	Correct	616	0.2203 ± 0.1934	20.3964 ± 2.2425	0.1599 ± 0.3078
SEED V	Incorrect	104	0.3901 ± 0.2044	19.9188 ± 2.1432	0.2784 ± 0.4467
SEED VII	Correct	1,160	0.2698 ± 0.2014	24.6769 ± 2.6439	0.1790 ± 0.3215
SEED VII	Incorrect	440	0.4138 ± 0.1839	24.8005 ± 2.9238	0.2469 ± 0.3777

TABLE XI

EXPLORATORY SUBJECT-PAIRED ACCURACY ANALYSIS FOR THREE VARIANTS. MEAN DIFFERENCES (FULL MINUS VARIANT) ARE IN PERCENTAGE POINTS AND USE THE MAIN ACCURACY TABLES. ADJUSTED P-VALUES AND PAIRED COHEN’S EFFECT SIZES d_z ARE DERIVED FROM THE SUBJECT-FOLD ANALYSIS. HOLM ADJUSTMENT COVERS 12 COMPARISONS WITHIN EACH TEST FAMILY; SUBJECT-LEVEL ACCURACY IS FIRST AVERAGED OVER FIVE SEEDS WITHIN EACH FOLD. THE ANALYSIS DOES NOT SEPARATELY TEST VARIABILITY ACROSS SEEDS.

Dataset	Variant	Subjects	Δ (pp)	Paired t , p_{Holm}	Wilcoxon, p_{Holm}	d_z
SEED	w/o Spectral Conditioning	15	16.45	9.48×10^{-6}	0.000366	1.970
SEED	w/o Quantum Branch	15	6.82	8.84×10^{-5}	0.000945	1.403
SEED	Quantum-Only Head	15	8.89	1.07×10^{-5}	0.000366	1.901
SEED IV	w/o Spectral Conditioning	15	21.86	4.19×10^{-10}	0.000366	4.670
SEED IV	w/o Quantum Branch	15	10.93	1.28×10^{-7}	0.000366	2.863
SEED IV	Quantum-Only Head	15	16.02	1.25×10^{-8}	0.000366	3.480
SEED V	w/o Spectral Conditioning	16	34.73	5.41×10^{-11}	0.000275	4.849
SEED V	w/o Quantum Branch	16	17.78	2.57×10^{-5}	0.000275	1.589
SEED V	Quantum-Only Head	16	30.56	4.20×10^{-10}	0.000275	4.146
SEED VII	w/o Spectral Conditioning	20	24.81	1.16×10^{-11}	2.29×10^{-5}	3.701
SEED VII	w/o Quantum Branch	20	12.75	1.45×10^{-9}	2.29×10^{-5}	2.741
SEED VII	Quantum-Only Head	20	19.62	1.63×10^{-9}	2.29×10^{-5}	2.700

TABLE XII

STORED MODEL SIZE AND PARTIAL PROFILER COUNTS FOR FULL QUANKAN WITH INPUT [1, 32, 62, 5]. FP32 SIZE INCLUDES WEIGHTS ONLY; MACS AND FLOPS ARE REPORTED IN MILLIONS AND EXCLUDE UNPROFIED OPERATIONS AND QUANTUM SIMULATION.

Dataset	Parameters	FP32 weights (MiB)	MACs (M)	FLOPs (M)
SEED	589,366	2.25	20.10	40.19
SEED IV	614,424	2.34	20.10	40.20
SEED V	639,739	2.44	20.11	40.21
SEED VII	691,652	2.64	20.12	40.24

The gate analysis is consistent with stronger residual weighting when the classical prediction is less confident. However, the effective correction also depends on the learned fusion coefficient and the feature-residual magnitude; SEED illustrates that a larger gate need not imply a larger mean correction. The bounded coefficient limits the scaling factor, not the norm of the unconstrained feature residual. These distinctions separate the implemented mechanism from stronger

claims about calibration, guaranteed error correction, or quantum advantage.

C. Efficiency and Limitations

The stored parameter budget and FP32 weight footprint establish a compact model size within the evaluated implementations. They do not establish a superior accuracy-latency trade-off against external baselines, whose costs have not been measured under a common setup. Partial profiler counts and analytic state-vector estimates also cannot be substituted for measured end-to-end latency, peak memory, or physical-device execution cost. Modules retained after component suppression explain why some functional removals change predictions without changing stored parameter counts.

Three limits bound the conclusions. First, the literature comparison combines results from independently implemented studies, so preprocessing, optimization, and target-data access can affect comparability even within cross-subject evaluation. Second, subject-paired significance tests are exploratory be-

TABLE XIII

ANALYTIC STATE-VECTOR ARITHMETIC ESTIMATES PER SAMPLE, REPORTED SEPARATELY FROM THE CLASSICAL PROFILER. THE ONE-QUBIT COLUMN COUNTS LOGICAL GATE APPLICATIONS, TREATING EACH Rot AS ONE SIMULATED OPERATION. THESE ESTIMATES ARE NEITHER PHYSICAL GATE DECOMPOSITIONS NOR LATENCY MEASUREMENTS.

Circuit	Qubits	Layers	Amplitudes	One-qubit operations	Gate/readout FLOPs
Full (R_Y/R_Z /Rot; $Z/X/ZZ$)	4	4	16	48	11,300
w/o Rot	4	4	16	32	7,716
Z-only readout	4	4	16	48	10,924
1 layer	4	1	16	12	3,236
2 layers	4	2	16	24	5,924
3 layers	4	3	16	36	8,612
2 qubits	2	4	4	24	1,414
6 qubits	6	4	64	72	67,758
8 qubits	8	4	256	96	361,192

cause LOSO training sets overlap. Although each subject-level score averages five seeds, separate estimates of variability across seeds are not reported. Third, the quantum branch is evaluated by simulation, and neither physical-device performance nor computational advantage is demonstrated. The current evidence therefore supports the tested coordinated architecture under strict LOSO on the four SEED datasets, with broader generality requiring additional datasets and controlled computational benchmarks.

VII. CONCLUSION & FUTURE WORK

This work presented QuanKAN, a compact hybrid quantum–classical framework for cross-subject EEG emotion recognition. By combining spectrum-conditioned DE-LDS refinement, graph-aware spatio-temporal modeling, cross-subject representation regularization, and uncertainty-gated quantum residual learning, QuanKAN achieves strong generalization without relying on excessive model capacity. Under strict leave-one-subject-out evaluation, the model obtains accuracy/F1 scores of $87.41 \pm 6.05\%/87.12 \pm 6.64\%$, $78.43 \pm 4.48\%/78.12 \pm 4.67\%$, $86.81 \pm 3.72\%/86.60 \pm 3.95\%$, and $73.81 \pm 4.58\%/73.16 \pm 4.91\%$ on SEED, SEED IV, SEED V, and SEED VII, respectively. Ablations indicate complementary contributions across the architecture, with spectrum-conditioned input refinement producing the largest individual effect among the tested component removals. In particular, the weaker quantum-only configuration indicates that, among the tested configurations, the quantum pathway is most effective as a compact residual refinement of an established classical representation. Overall, the results demonstrate strong cross-subject performance within a comparatively compact sub-million-parameter architecture.

Future work will focus on strengthening this performance-efficiency trade-off and assessing its generality. We will first quantify parameter count, FLOPs, memory usage, and inference latency against representative lightweight and high-capacity EEG baselines, while exploring further compression through pruning and parameter sharing. Evaluation will also be extended beyond the SEED family to cross-dataset and

heterogeneous acquisition settings, accompanied by paired subject-level statistical analysis. Finally, because the present quantum pathway is evaluated using state-vector simulation, future studies will investigate finite-shot execution, realistic quantum noise, restricted connectivity, and physical quantum hardware to determine whether the uncertainty-gated residual formulation remains effective under practical deployment constraints.

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